



7358 - The Other Extreme: Enabling Characterization of Metal-poor Brown Dwarfs and testing our Understanding of Jupiter and Saturn

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

A population of metal-poor brown dwarfs, subdwarfs, has been identified over the last decade. Most of them are from our galaxy's thick disk and halo. Several JWST observations aim to constrain the composition of subdwarfs as they are essential for understanding low-mass star formation in the farthest reaches of our galaxy. However, we rely on theoretical models of subdwarf atmospheres to constrain their composition from spectra. While significant progress has been made in developing models for solar composition and metal-rich objects, driven by exoplanet science, very few models exist for metal-poor objects. The limited number of available models have ignored critical processes, like impact of atmospheric dynamics on non-C/N/O bearing gasses. Gasses like SiH₄ and GeH₄, long predicted in Jupiter, may play an outsized role in metal-poor objects, along with other H-dominated gasses. In Jupiter these molecules are all far from chemical equilibrium, suggesting that treating the impact of dynamics on these gasses will be crucial for subdwarfs. Therefore, we must develop models that include these processes and apply them to subdwarf atmospheres. Without such models, it will be very difficult to interpret the composition of subdwarfs from JWST observations. We will fill this major gap by developing a

modeling framework for subdwarfs that will include these critical processes. We will use this framework to create atmospheric model grids, which can be used to decipher their composition and constrain several uncertain atmospheric processes from JWST observations. Our results will allow stress-tests of our understanding of Jupiter with JWST by leveraging subdwarfs as laboratories.

OBSERVING DESCRIPTION